



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

FACULTY OF COMPUTING
UNIVERSITI TEKNOLOGI MALAYSIA

NETWORK COMMUNICATIONS

SECR1213-12

SEMESTER 1 2024/2025

PROJECT TASK 5: IP ADDRESSING SCHEME

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GROUP 5

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5.1 Get the Network Address from your lecturer

12.1	192.160.0.0/8
12.2	192.162.0.0/8
12.3	192.164.0.0/8
12.4	192.166.0.0/8
12.5	192.168.0.0/8
12.6	192.170.0.0/8

Table 5.1.1

Table 5.1.1 shows the list of network addresses of each group. The network address of our group is 192.170.0.0/8.

Network address details:

IP address (Decimal)	192.170.0.0
IP address (Binary)	11000000 10101010 00000000 00000000
Subnet Mask (Decimal)	255.0.0.0
Subnet Mask (Binary)	11111111 00000000 00000000 00000000

We partitioned the subnet mask into a network portion (8 bits) and a host portion (24 bits). By performing the AND operation of the IP address with the subnet mask, we identified the network address as 192.170.0.0. To calculate the broadcast address, the host portion is filled with all 1s, so the broadcast address is 192.170.255.255. The valid address range within our building spans from 192.170.0.1 to 192.170.255.254.

Network Address	192.0.0.0
Broadcast Address	192.255.255.255
Range of usable address	192.0.0.1 - 192.255.255.254

5.2 Network distribution

5.2.1 Network distribution for different work area

In order to improve internet speed and enhance the security of the faculty computing work areas, subnetting is done to divide the IP address space into fixed-size classes. Based on the network design requirements, we identified that we need 9 subnets to accommodate 9 different work areas, 4 bits ($2^4=16$) is borrowed from the host portion to construct a subnet IP address. The table below lists each subnet network portion and host portion and subnet distribution for each area.

192.170.0.0/8 → 1100 0000 . 1010 1010 . 0000 0000 . 0000 0000

(Borrow 4 bits from host to define 9 subnets)

The subnet mask for each subnet is /12 now.

Subnet	Network Portion	Subnet bits	Host Portion	IP address	
0	11000000	0000	0000 00000000 00000000	Network Address	192.0.0.0
			1111 11111111 11111111	Broadcast Address	192.15.255.255
1	11000000	0001	0000 00000000 00000000	Network Address	192.16.0.0
			1111 11111111 11111111	Broadcast Address	192.31.255.255
2	11000000	0010	0000 00000000 00000000	Network Address	192.32.0.0
			1111 11111111 11111111	Broadcast Address	192.47.255.255

3	11000000	0011	0000 00000000 00000000	Network Address	192.48.0.0
			1111 11111111 11111111	Broadcast Address	192.63.255.255
4	11000000	0100	0000 00000000 00000000	Network Address	192.64.0.0
			1111 11111111 11111111	Broadcast Address	192.79.255.255
5	11000000	0101	0000 00000000 00000000	Network Address	192.80.0.0
			1111 11111111 11111111	Broadcast Address	192.95.255.255
6	11000000	0110	0000 00000000 00000000	Network Address	192.96.0.0
			1111 11111111 11111111	Broadcast Address	192.111.255.255
7	11000000	0111	0000 00000000 00000000	Network Address	192.112.0.0
			1111 11111111 11111111	Broadcast Address	192.127.255.255
8	11000000	1000	0000 00000000 00000000	Network Address	192.128.0.0
			1111 11111111	Broadcast Address	192.143.255.255

			11111111		
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Distribution of subnet for different work area:

Subnet	Work Area	Network Address	Broadcast Address	Range of usable Address	Subnet Mask Address
Ground Floor					
0	Embedded Lab	192.0.0.0	192.15.255.255	192.0.0.1 - 192.15.255.254	/12 255.240.0.0
1	Cisco Network Lab	192.16.10.0	192.31.255.255	192.16.0.1 - 192.31.255.254	/12 255.240.0.0
2	General Purpose Lab1	192.32.0.0	192.47.255.255	192.32.0.1 - 192.47.255.254	/12 255.240.0.0
3	General Purpose Lab2	192.48.0.0	192.63.255.255	192.48.0.1 - 192.63.255.254	/12 255.240.0.0
First Floor					
4	Video Conferencing Room	192.64.0.0	192.79.255.255	192.64.0.1- 192.79.255.254	/12 255.240.0.0
5	Server Room	192.80.0.0	192.95.255.255	192.80.0.1- 192.95.255.254	/12 255.240.0.0
6	Student Lounge	192.96.0.0	192.111.255.255	192.96.0.1- 192.111.255.254	/12 255.240.0.0
7	Hybrid Classroom	192.112.0.0	192.127.255.255	192.112.0.1- 192.127.255.254	/12 255.240.0.0
8	Academic Office and Staff Meeting Room	192.128.0.0	192.143.255.255	192.128.0.1- 192.143.255.254	/12 255.240.0.0

5.2.2 Network distribution for workstation in different work areas.

PC	Personal Computer
PJ	Projector
SV	Server
WAP	Wireless Access Point
SMB	Smartboard

First Floor:

1. General Lab 1

Devices	IP Address
PC1	192.32.0.1
PC2	192.32.0.2
PC3	192.32.0.3
PC4	192.32.0.4
PC5	192.32.0.5
PC6	192.32.0.6
PC7	192.32.0.7
PC8	192.32.0.8
PC9	192.32.0.9
PC10	192.32.0.10
PC11	192.32.0.11
PC12	192.32.0.12
PC13	192.32.0.13
PC14	192.32.0.14
PC15	192.32.0.15
PC16	192.32.0.16
PC17	192.32.0.17
PC18	192.32.0.18

PC19	192.32.0.19
PC20	192.32.0.20
PC21	192.32.0.21
PC22	192.32.0.22
PC23	192.32.0.23
PC24	192.32.0.24
PC25	192.32.0.25
PC26	192.32.0.26
PC27	192.32.0.27
PC28	192.32.0.28
PC29	192.32.0.29
PC30	192.32.0.30
PC31	192.32.0.31
PC32	192.32.0.32
WAP1	192.32.0.33
PJ1	192.32.0.34

2. General Lab 2

Devices	IP Address
PC33	192.48.0.1
PC34	192.48.0.2
PC35	192.48.0.3
PC36	192.48.0.4
PC37	192.48.0.5
PC38	192.48.0.6
PC39	192.48.0.7
PC40	192.48.0.8
PC41	192.48.0.9

PC42	192.48.0.10
PC43	192.48.0.11
PC44	192.48.0.12
PC45	192.48.0.13
PC46	192.48.0.14
PC47	192.48.0.15
PC48	192.48.0.16
PC49	192.48.0.17
PC50	192.48.0.18
PC51	192.48.0.19
PC52	192.48.0.20
PC53	192.48.0.21
PC54	192.48.0.22
PC55	192.48.0.23
PC56	192.48.0.24
PC57	192.48.0.25
PC58	192.48.0.26
PC59	192.48.0.27
PC60	192.48.0.28
PC61	192.48.0.29
PC62	192.48.0.30
PC63	192.48.0.31
PC64	192.48.0.32
WAP2	192.48.0.33
PJ2	192.48.0.34

3. Cisco Network Lab

Devices	IP Address
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PC65	192.16.0.1
PC66	192.16.0.2
PC67	192.16.0.3
PC68	192.16.0.4
PC69	192.16.0.5
PC70	192.16.0.6
PC71	192.16.0.7
PC72	192.16.0.8
PC73	192.16.0.9
PC74	192.16.0.10
PC75	192.16.0.11
PC76	192.16.0.12
PC77	192.16.0.13
PC78	192.16.0.14
PC79	192.16.0.15
PC80	192.16.0.16
PC81	192.16.0.17
PC82	192.16.0.18
PC83	192.16.0.19
PC84	192.16.0.20
PC85	192.16.0.21
PC86	192.16.0.22
PC87	192.16.0.23
PC88	192.16.0.24
PC89	192.16.0.25
PC90	192.16.0.26
PC91	192.16.0.27

PC92	192.16.0.28
PC93	192.16.0.29
PC94	192.16.0.30
WAP3	192.16.0.31
SMB1	192.16.0.32

4. Embedded Lab

Devices	IP Address
PC95	192.0.0.1
PC96	192.0.0.2
PC97	192.0.0.3
PC98	192.0.0.4
PC99	192.0.0.5
PC100	192.0.0.6
PC101	192.0.0.7
PC102	192.0.0.8
PC103	192.0.0.9
PC104	192.0.0.10
PC105	192.0.0.11
PC106	192.0.0.12
PC107	192.0.0.13
PC108	192.0.0.14
PC109	192.0.0.15
PC110	192.0.0.16
PC111	192.0.0.17
PC112	192.0.0.18
PC113	192.0.0.19
PC114	192.0.0.20

PC115	192.0.0.21
PC116	192.0.0.22
PC117	192.0.0.23
PC118	192.0.0.24
PC119	192.0.0.25
PC120	192.0.0.26
PC121	192.0.0.27
PC122	192.0.0.28
PC123	192.0.0.29
PC124	192.0.0.30
PC125	192.0.0.31
PC126	192.0.0.32
PC127	192.0.0.33
WAP4	192.0.0.34
PJ3	192.0.0.35
SMB2	192.0.0.36

Second Floor:

1. Video Conferencing Room

Devices	IP Address
PC128	192.64.0.1
PC129	192.64.0.2
SMB3	192.64.0.3
WAP5	192.64.0.4

2. Server Room

Devices	IP Address
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PC130	192.80.0.1
SV1	192.80.0.2
SV2	192.80.0.3
SV3	192.80.0.4
SV4	192.80.0.5
SV5	192.80.0.6
SV6	192.80.0.7

3. Student Lounge

Devices	IP Address
PC131	192.96.0.1
PC132	192.96.0.2
WAP6	192.96.0.3

4. Hybrid Classroom

Devices	IP Address
PC133	192.112.0.1
PC134	192.112.0.2
PC135	192.112.0.3
PC136	192.112.0.4
PC137	192.112.0.5
PC138	192.112.0.6
PC139	192.112.0.7
PC140	192.112.0.8
PC141	192.112.0.9
PC142	192.112.0.10
PC143	192.112.0.11
PC144	192.112.0.12

PC145	192.112.0.13
PC146	192.112.0.14
PC147	192.112.0.15
PC148	192.112.0.16
PC149	192.112.0.17
PC150	192.112.0.18
PC151	192.112.0.19
PC152	192.112.0.20
PC153	192.112.0.21
PC154	192.112.0.22
PC155	192.112.0.23
PC156	192.112.0.24
PC157	192.112.0.25
PC158	192.112.0.26
PC159	192.112.0.27
PC160	192.112.0.28
PC161	192.112.0.29
PC162	192.112.0.30
PC163	192.112.0.31
PC164	192.112.0.32
WAP7	192.112.0.33
PJ4	192.112.0.34

5. Academic Office

Devices	IP Address
PC165	192.128.0.1
PC166	192.128.0.2
PC167	192.128.0.3

PC168	192.128.0.4
PC169	192.128.0.5
PC170	192.128.0.6
PC171	192.128.0.7
PC172	192.128.0.8
PC173	192.128.0.9
PC174	192.128.0.10
PC175	192.128.0.11
PC176	192.128.0.12
PC177	192.128.0.13
PC178	192.128.0.14
PC179	192.128.0.15
PC180	192.128.0.16
PC181	192.128.0.17
PC182	192.128.0.18
PC183	192.128.0.19
PC184	192.128.0.20
PC185	192.128.0.21
PC186	192.128.0.22
WAP8	192.128.0.23

Meeting Minutes #1

Date/Time		15 Jan 2024 2:00pm	
Location		Online (Google Meet)	
Agenda		1. Discuss the question on Task 5	
		2. Discuss report format	
		3. Task Separation and Distribution	
Meeting Host		Lim Yu Han	
Attendance			
Name		Time	Reason For Absence
Goh Jiale		2:00pm	-
Joanne Ching Yin Xuan		2:00pm	-
Evelyn Goh Yuan Qi		2:00pm	-
Lim Yu Han		2:00pm	-
Minutes			
No.	Item Discussed	Details	Person-In-Charge
1	Discuss of question on Task 5	Lim Yu Han explains Task 5 deliverables.	Lim Yu Han
		All members discuss the task and select topics to do.	All Member
2	Discuss Report Format	Goh Jiale and Joanne list down the Task 5 deliverables.	All Member
		All members decide to include calculation for network distribution in the report.	
3	Task Distribution	Get IP address from the lecturer and calculate IP address and subnet mask for the project.	Evelyn
		Calculate the network distribution for each subnet including calculation.	Joanne & Lim Yu Han

		Distribute the network for each workstation	Goh Jiale
4	Meeting ended		